

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2024-Present	Doctor of Philosophy	Indian Institute of Technology Kanpur	9.63*/10
2022-2024	M.Sc., Applied Geology	Presidency University, Kolkata	8.45/10
2019-2022	B.Sc., Geology	Presidency University, Kolkata	8.88/10
2019	ISC [XII]	Calcutta Boys' School, Kolkata	89%
2017	ICSE [X]	Calcutta Boys' School, Kolkata	92.8%

Ph.D. Thesis

Thesis Work | Supervisor: Prof. Tajdarul Hassan Syed| Dept. of Earth Sciences, IIT Kanpur (Jul'24-Ongoing)

- Title: Characterizing Permafrost and Active Layer Dynamics in the Ladakh Himalaya using Satellite and Ground-based Observations
- Aim: The aim of this work is to characterize permafrost distribution and active layer dynamics in the Western Himalaya using integrated InSAR and PolSAR-based remote sensing approaches.

M.Sc. Dissertation

Thesis Work | Supervisor: Dr. Aditya Sarkar| Dept. of Geology, Presidency University, Kolkata (Jul'23-Jul'24)

- Title: Appraisal of Hydrogeological Variability in Parts of Greater Kolkata in Perspective of Urbanization
- Aim: The aim of this work was to evaluate the impact of rapid urbanization on groundwater dynamics, hydrochemistry, and vulnerability in parts of the Greater Kolkata region using integrated hydrogeological, geochemical, and geospatial analyses.

Publications

Majumder, S.,Mukherjee, S., Sarkar, A. et al. Identification of flood hazard-prone coastal zones using remote sensing tools: a study from the Sundarbans Deltaic Region of West Bengal, India. Proc.Indian Natl. Sci. Acad. (2025). <https://doi.org/10.1007/s43538-025-00681-y>

Conference Proceedings

- Poster Presentation - Insights on the Hydrological Regime of Higher Himalayan Catchments using Hydrochemistry and Stable Hydrogen and Oxygen Isotope Compositions of River Water and Groundwater |(Conference on Integrated Earth (CITE-2026), IISER Pune) (Sept'26)
- Poster Presentation - Delineation of Groundwater Vulnerability Zones using DRASTIC Model in the Alluvial and West Marginal Tract of West Bengal, India |(Sustain Istanbul 2024, Turkey) (Oct'24)
- Oral Presentation - Assessing the impact of anthropogenic activities linked to seawater intrusion in coastal aquifers: A case study in the Balasore District, Odisha, India |(CGWA, Kolkata) (Jan'24)
- Poster Presentation - Evaluation of Saltwater Ingression in Freshwater Resources Under the Influence of Tropical Cyclones in Coastal Aquifers of Eastern Coast, India |(IISWBM, Kolkata) (Apr'23)
- Poster Presentation - Correlating Hydrogeological Characteristics with Climate of Birbhum and Murshidabad Districts, West Bengal |(TROPMET-2022, IISER Bhopal, M.P.) (Nov'22- Dec'22)

Trainings

- Summer Training - Well hydraulics and Pumping test data analysis |(RGNGWTRI, Raipur, C.G.) (Jul'23)
- Industrial Field Training - Structural mapping of Singhbhum Shear Zone in Kumari-Ukam-Kudada area |(AMD-ER/DAE, Jamshedpur, Jharkhand) (Jun'23)
- Summer Training - Depicting hydrometeorological characteristics of the central part of West Bengal by correlating satellite imagery with regional hydrodynamics |(Presidency University, Kolkata) (Jul'22- Sept'22)

Technical Skills

- Languages: Python, R, LaTeX
- Libraries: NumPy, SciPy, Pandas, Matplotlib

- **Utilities:** MS Office suite, Google Earth Engine, QGIS, ArcGIS, ENVI, GMT, Modflow, Surfer, Rockworks, Grapher, Origin Pro

Position of Responsibility

- **Sports Secretary** | Earth Science Society, IIT Kanpur (Jul'25-Jul'26)
- **President** | Earth Science Society, IIT Kanpur (Jul'24-Jul'25)
- **Publication Secretary** | Geological Institute, Presidency University, Kolkata (Jul'23-Jul'24)
- **Vice President** | SPE Student Chapter, Presidency University, Kolkata (Jul'22-Jul'23)
- **Social Activities Chair** | SPE Student Chapter, Presidency University, Kolkata (Jul'21-Jul'22)
- **Sports Captain** | United Students Organization(USO), Calcutta Boys' School, Kolkata (Apr'18-Apr'19)

Achievements & Awards

- Awarded the prestigious **TCS Research Scholar Program (Cycle 20) Scholarship** in recognition of academic and research potential, providing financial and academic support for doctoral research (Jul'2026)
- Qualified **GATE Exam twice (2023 and 2024)**, demonstrating consistent academic excellence.
- **Second Prize** - Oral Presentation on Evaluation of Saltwater Ingression in Freshwater Resources Under the Influence of Tropical Cyclones in Coastal Aquifers of Eastern Coast, India at IISWBM, Kolkata (Apr'2023)

Extra-Curricular activities

- Secured **1st position** in instiSMASH'2026- Badminton tournament(Doubles), organized by IIT Kanpur (Jan'2026)
- Represented **IIT Kanpur** at Inter-IIT Sports Meet'2025 in Athletics, organized by IIT Hyderabad (Dec'2025)
- Secured **1st position** in Raquethalon'2025- Badminton tournament, organized by IIT Kanpur (Aug'2025)
- Secured **1st position** in Chiddi Balla'2025- Badminton tournament(Doubles), organized by IIT Kanpur (Jun'2025)
- Secured **2nd position** in Chiddi Balla'2025- Badminton tournament(Singles), organized by IIT Kanpur (Jun'2025)
- Secured **1st position** in Sportech'2025 in 4x100m Relay race, organized by IIT Delhi (Mar'2025)